Data Flow Diagram

Sizland Land Acquisition System

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Abstract

This document illustrates the flow of data within the Sizland Land Acquisition System, highlighting the interactions between external entities, core processes, and data stores. The system utilizes a notary-mediated escrow model to ensure security and legal compliance throughout the land purchase lifecycle.

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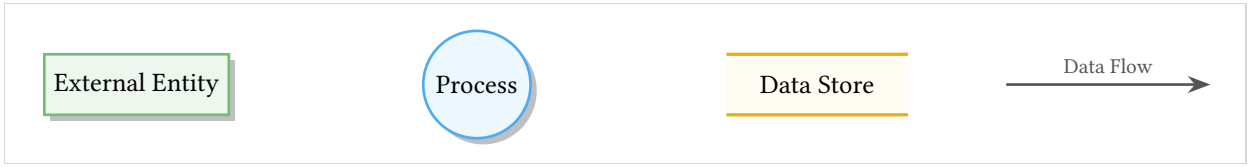
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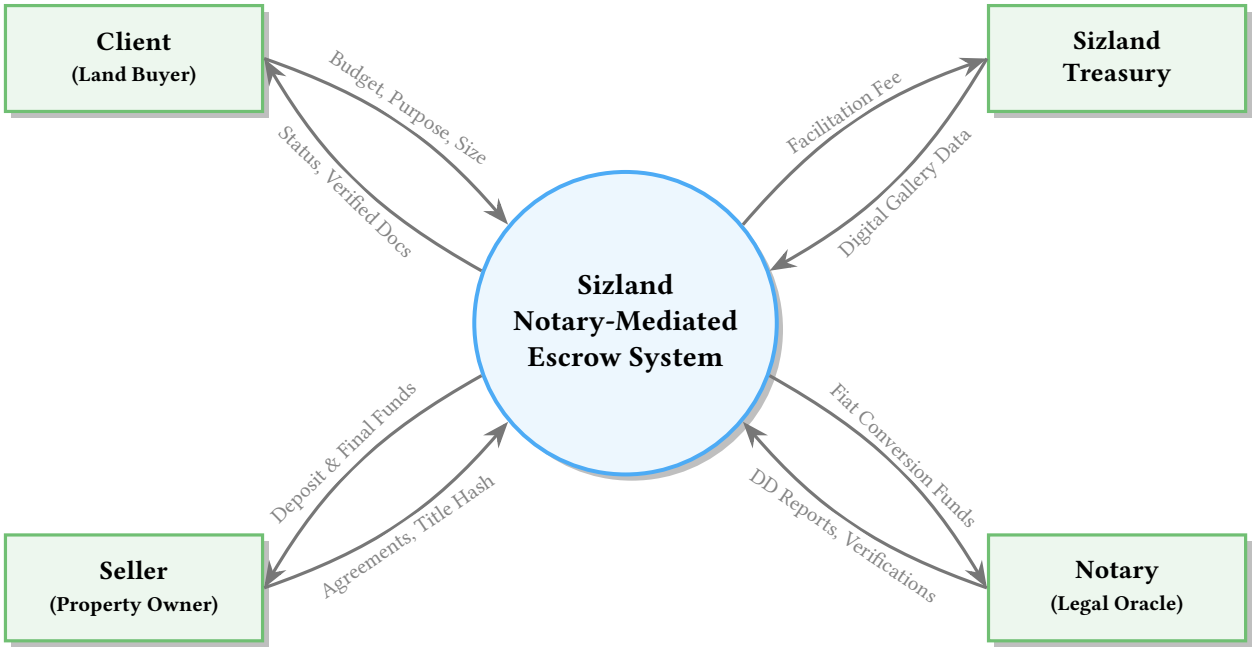
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DFD Notation Architecture



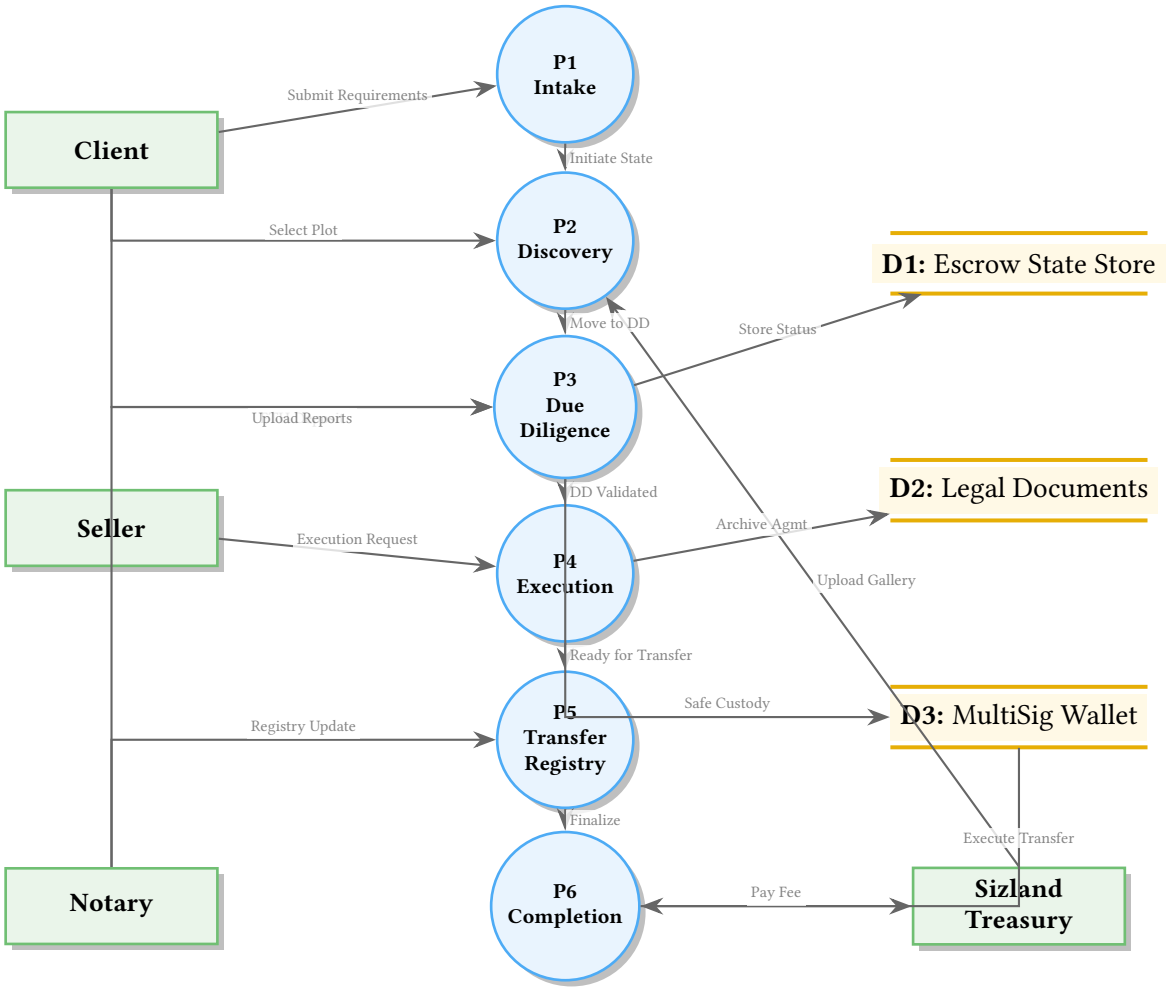
Level 0 — Context Diagram



Information

The Context Diagram represents the highest level of abstraction, showing individual entities interacting with the core system as a single process.

Level 1 — Decomposed Data Flow Diagram



Process Catalog

P1: Intake Client provides initial parameters (budget, size, purpose). System initializes a new escrow transaction state in **D1**.

P2: Discovery Sizland Treasury uploads property gallery. Client reviews and selects a specific plot, updating the escrow state.

P3: Due Diligence Client funds the escrow. Notary uploads the DD report and search cert. Client verifies these via the UI. Funds are held in **D3**.

P4: Execution Notary sets rates. Both parties provide cryptographic signatures on the sale agreement, which is stored in **D2**.

P5: Transfer Notary uploads the lodgement receipt once the transfer is initiated at the lands registry.

P6: Completion Final title documents are hashed and uploaded. Multi-sig releases funds to the seller and the facilitation fee to the treasury.

Data Architecture

D1: Escrow State Store (On-Chain Mapping)

Attribute	Type	Purpose
client / seller	address	Participants' wallet addresses
notary	address	Designated legal oracle
state	enum	Current workflow stage (Intake → Completed)
depositedAmount	uint256	Real-time balance held in escrow
facilitationBP	uint16	Fee percentage in basis points

D2: Legal Documents (IPFS + Hashes)

Attribute	Type	Purpose
documentHash	bytes32	Verifiable IPFS hash of the PDF document
verifiedBy	address	Address of the party who validated the document
signingStatus	enum	Tracking for multi-party signatures

D3: MultiSig Wallet (Asset Vault)

The MultiSig is a dedicated contract (`MultiSigWallet.sol`) that acts as the physical vault for ETH assets. It only releases funds when the required number of owners (or the `trustedCaller`) execute a transaction.

Sizland Land Acquisition Infrastructure
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